

Daily Problem: 17–Feb–2026

Problem Statement

Show that any language accepted by a finite-state automaton with an *array* is regular, assuming the array length is a fixed constant independent of the input length.

More precisely: consider a machine that has a finite control (like a DFA) and, in addition, an array of length m for some fixed constant m . Each array cell stores a symbol from a fixed finite alphabet Γ . The machine may read and update the array during its computation. Prove that any language recognized by such a machine is regular.

Proof

Let Σ be the input alphabet. Fix an array length $m \geq 0$ (a constant, not depending on input size), and a finite array alphabet Γ . Consider any finite-state automaton with such an array; call it A .

Step 1: The array gives only finitely many configurations

Because the array has fixed length m and each cell stores a symbol from the finite set Γ , the number of possible array contents is

$$|\Gamma|^m,$$

which is finite.

Let Q be the finite set of control states of A . Then a *global configuration* of the machine (ignoring the input head position) is completely determined by:

- the current control state $q \in Q$, and
- the current array content $s \in \Gamma^m$.

Hence the total number of possible pairs (q, s) is

$$|Q| \cdot |\Gamma|^m,$$

which is finite.

Step 2: Build an equivalent DFA by “folding” the array into the state

We now construct a DFA D that simulates A exactly, by storing the entire array content inside its state.

Define

$$Q' = Q \times \Gamma^m.$$

A state $(q, s) \in Q'$ means: “ A is currently in control state q and the array currently contains s .”

Start state. Let $s_0 \in \Gamma^m$ be the initial array content of A . Let $q_0 \in Q$ be the initial control state. Then set

$$q'_0 = (q_0, s_0).$$

Transition function. Fix an input symbol $a \in \Sigma$. Suppose the machine A , when in control state q with array content s , reads a and deterministically updates:

- the control state to q^* , and
- the array content to s^* .

(Any model of an “array automaton” specifies such a deterministic next-step update rule based on the current control state, the current input symbol, and the current array content.) Define the DFA transition:

$$\delta'((q, s), a) = (q^*, s^*).$$

Because A ’s update rule is deterministic, δ' is well-defined and is a function.

Accepting states. Let $F \subseteq Q \times \Gamma^m$ be the set of accepting configurations of A , i.e. those pairs (q, s) for which A would accept after consuming the entire input. Define

$$F' = F \subseteq Q'.$$

This completes the definition of the DFA

$$D = (Q', \Sigma, \delta', q'_0, F').$$

Step 3: Correctness

We prove by induction on the length of the input prefix that D simulates A exactly.

Let $w \in \Sigma^*$, and write $w = a_1 a_2 \cdots a_n$. Let (q_i, s_i) be the configuration of A after reading the first i symbols of w , and let (q'_i, s'_i) be the state of D after reading the same prefix. We show that for all $i = 0, 1, \dots, n$,

$$(q'_i, s'_i) = (q_i, s_i).$$

This is true at $i = 0$ by construction of the start state. If it holds for i , then when reading a_{i+1} , both A and D apply the same deterministic update rule from (q_i, s_i) , so they move to the same next pair. Thus the claim holds for all i by induction.

Therefore D accepts exactly the same inputs as A , i.e.

$$L(D) = L(A).$$

Step 4: Conclude regularity

Since D is a DFA and recognizes $L(A)$, the language $L(A)$ is regular.

Conclusion

Because the array has *fixed constant length* and stores symbols from a finite alphabet, the array contributes only finitely many possible memory configurations. We can fold that finite memory into the state space of a DFA. Hence every language recognized by a finite-state automaton with a fixed-length array is regular.